Project name: Talk to Character

Tracks: Conversation as the Interface (primary); Put a Face to the Voice (secondary); Agents That Act (supporting tool use).

One-sentence description: Pause a movie and talk face to face with its characters in their own voices, choosing whether they know the ending or only what has happened so far.

Project write-up

Sometimes the person you most want to ask about a movie is someone inside it. Why did you lie? Do you trust her? What are you afraid of right now? Searching for answers can spoil the story, while a character bot that knows the ending misses what makes those questions interesting. Talk to Character lets movie lovers pause a prepared film, select a face on screen, and start a conversation. Higgs Realtime supports multilingual voice conversations and interruptions, can clone voices preserve the character's sound, while Higgs Avatar brings their face into the exchange. The character's perspective changes with the pause point. We turn subtitles and screenplay information into scene memories, evolving relationships, and emotional context, then limit what the character can access to what they have experienced so far. Tools retrieve relevant memories and relationships during conversation, and a separate spoiler checker helps catch future plot details before they reach the viewer. Ask the same character the same question at two different moments, and their answer can change. After watching, an optional omniscient mode opens up conversations about the ending and "what if" questions.

How we built it

We built Talk to Character with React Frontend and FastAPI Backend. Higgs Realtime 1.5 Preview generates character replies directly as speech, with multilingual conversations and interruption handling. Higgs STT 3.1 transcribes the user's speech within the session. We register character voice clones through Boson's voice API using extracted dialogue clips. The backend also supports Higgs TTS 3 for text-to-speech requests. In avatar mode, Higgs Avatar animates a face from the paused scene using the generated reply audio, with cached listening animations between turns. The hardest part was keeping the character's knowledge consistent with the pause point. We process subtitles and screenplay information into scene memories and character states, then filter them by what the viewer has seen and what the character experienced. For nonlinear films, we also check story chronology. Relevant memories and relationship context are attached to each question before the model responds. A separate, stable Higgs Realtime model checks response sentences for future plot details. The frontend can request a safe replacement or cut a reply short. Voice mode coordinates these

checks with buffered audio and interruptions; avatar mode waits for the checks before rendering.

Team Information

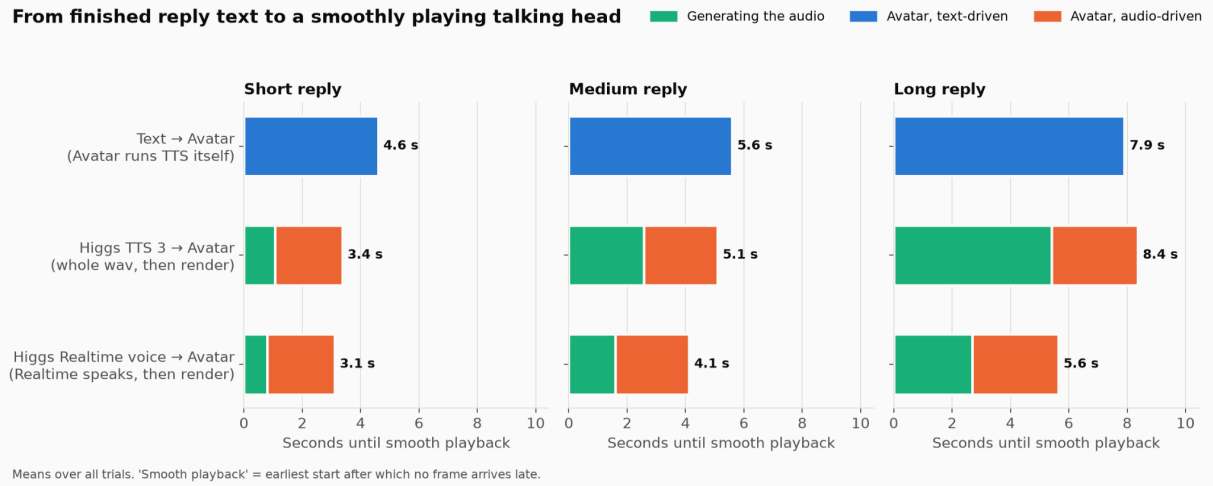
Members:

- Haochen Ding (@Dennis-Ding1): Backend development, Higgs API integration, and character knowledge processing.
- Rolland He (@Rolland-He): Real-time voice interaction, avatar integration, and testing.
- Yinuo Yang (@CiaraYang): Frontend interface and user experience, including the movie player, character selection, and conversation panel.

Repository: https://github.com/Dennis-Ding1/Talk_to_Character.git

Known limitations:

- Avatar mode adds latency because speech must be generated, checked for spoilers, and rendered into video before playback. In our tests, Realtime audio-to-avatar took roughly 3–6 seconds to reach smooth playback, while sending text directly to Avatar took around 5–8 seconds. Longer replies generally take more time.



- Face detection and avatar generation are less reliable for anime characters and heavily stylized faces. Clearly visible eyes, a nose, and a mouth are needed; exaggerated proportions or artistic effects can cause detection failures or poor animation.